

E-scooters far more dangerous than motorbikes or cycles, analysis suggests

Study of accidents in England and Wales shows e-scooter riders three times more likely to suffer traumatic brain injury than bikers



The study also showed e-scooter riders sustained more internal organ and blood vessel damage than motorcyclists or pedal cyclists. Photograph: Mike Kemp/In Pictures/Getty Images

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Electric scooter riders are three times more likely to suffer traumatic brain injuries than motorcyclists, according to an analysis of accidents in [England](#) and Wales – which found women were particularly at risk of serious harm.

Brain injuries accounted for more than a third of adult injuries on e-scooters, but riders also sustained more internal organ and blood vessel damage than did motorcyclists or pedal cyclists, the study found.

The government has backed trials of rental e-scooters in England since 2020, but an additional 1m or so e-scooters are privately owned in the UK. These cannot legally be used on public roads, paths or cycle lanes, but police forces across the country have confiscated thousands because their owners were breaking the law.

The accident analysis reveals that women were involved in about a third more collisions than men – and were more than twice as likely to report severe injuries. Children were also at risk with, 16% of e-scooter injuries in under 18s.

The findings have led to calls to make e-scooters more stable and safer, especially for women who suffer more chest injuries because the handlebars are typically set at a height for men. Mandatory safety training and making helmets compulsory would also help, experts said.

David Bodansky, a consultant hand, trauma and orthopaedic surgeon at Chelsea and Westminster hospital NHS foundation trust, turned to the national major trauma registry to analyse injuries from e-scooter, motorbike and bicycle collisions in 18 cities in England and [Wales](#) between 2020 and 2022.

He found e-scooter riders were more than three times as likely to suffer traumatic brain injury or blood vessel damage and 1.5 times more likely to injure internal organs than motorbike riders, and they fared worse than cyclists for these injuries. Their risk of bone fractures was lower, however, than cyclists or motorcyclists.

The data shed no light on why e-scooters are involved in so many accidents, but Bodansky believes that design is a factor. “They’ve got very small wheels and it’s very easy to jam one of those into a pothole. You’ll then go head over heels over the front and land on your head,” he said. Less than 6% of e-scooter riders reported wearing a helmet, according to the study, in [Scientific Reports](#).

While some [previous work](#) found women were more likely to ride e-scooters on the pavement, other research saw no sex differences. In a [2024 study](#), Dr Petya Ventsislavova, senior lecturer in transport psychology at Nottingham Trent University, found more than half of e-scooter riders rode on pavements, and with another person at the same time, while men reported using their phones and shooting red lights more than did women.

“These figures highlight the urgent need for comprehensive legislation governing private e-scooters,” said Ventsislavova. The legislation should clarify not only where and how e-scooters can be used but also include mandatory safety training, she said.

“Unlike cars and motorcycles, e-scooters were introduced without any requirement for formal training, despite being motorised vehicles,” she said. “E-scooters handle very differently from bicycles. Riders stand upright, have a higher centre of gravity, use more sensitive steering, and are more vulnerable to sudden stops caused by small wheels striking kerbs, potholes or road debris. These characteristics make them inherently less stable and contribute to the severity of injuries when crashes occur.”

A Department for Transport spokesperson said: “These are concerning findings, and our thoughts are with anyone who has been injured on our roads. [Road safety](#) is our top priority, which is why our e-scooter trials are testing their use in a controlled environment now, before updating the law to ensure they can be used safely more widely.”

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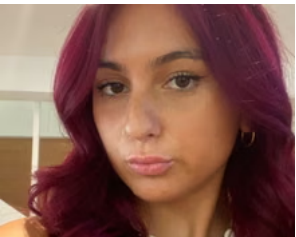
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